The binary DirtNet task became a 10-class detector

10 classes

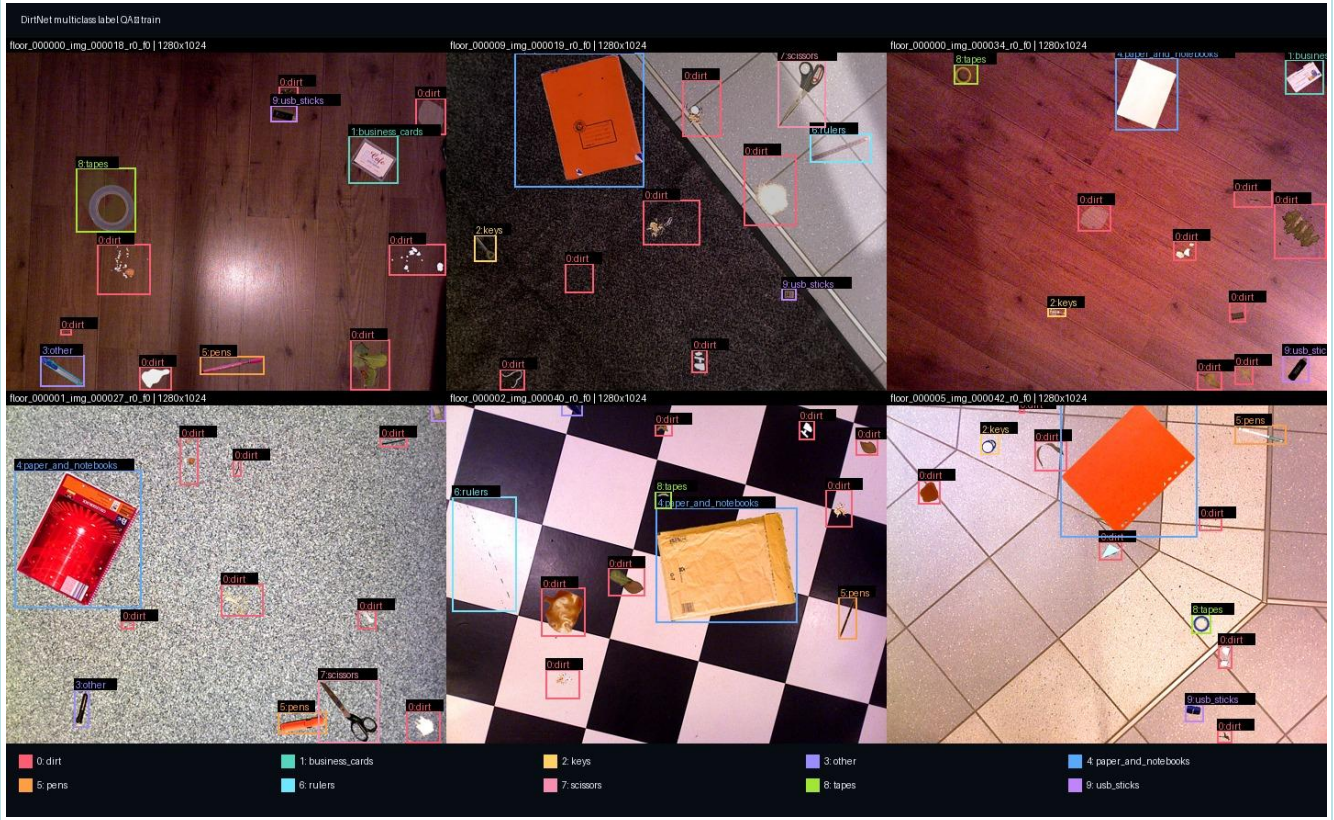
dirt + 9 office-object categories

Dataset

860 training images · 8,626 boxes
296 validation images · 2,962 boxes
80 held-out real images · 644 boxes

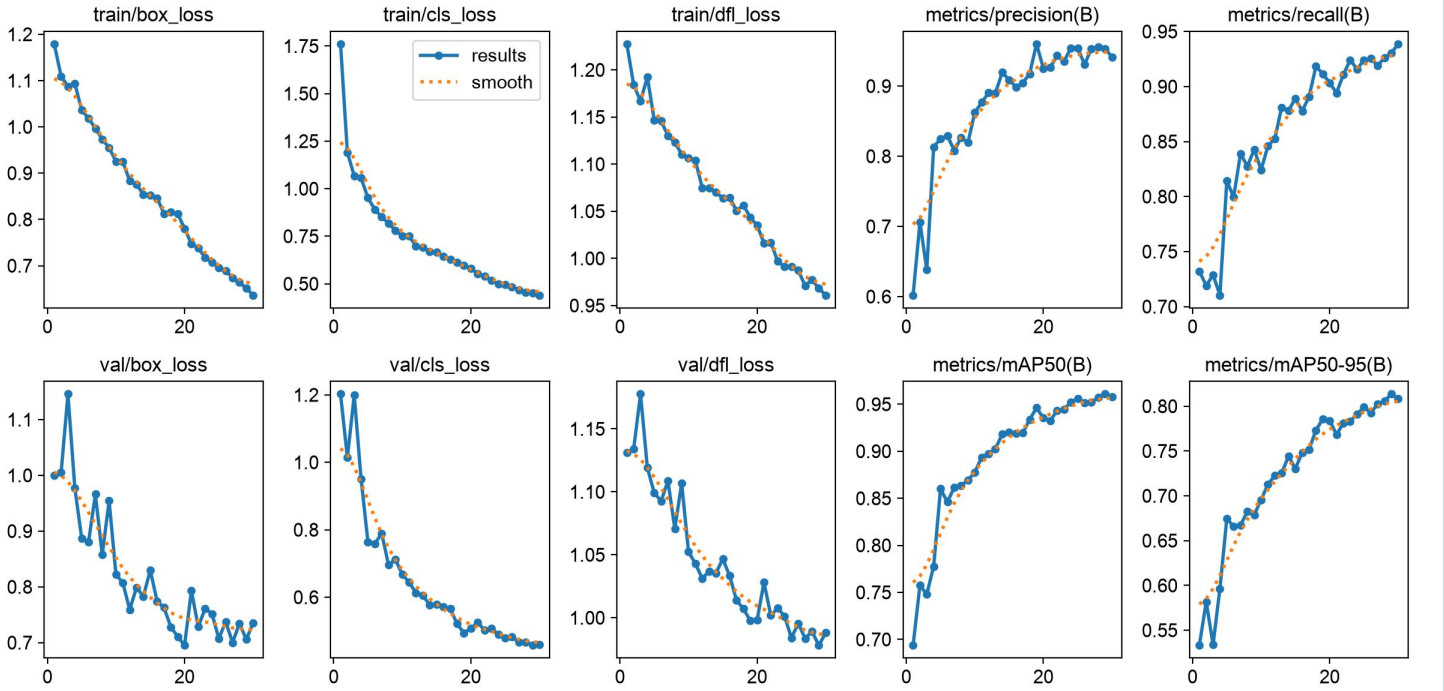
Controlled baseline

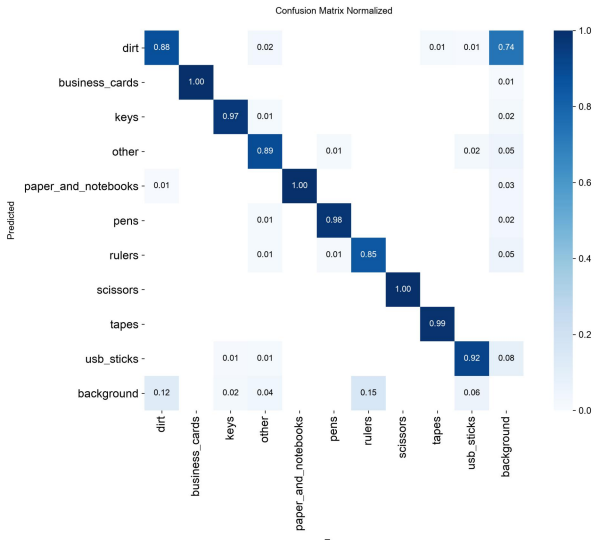
YOLO11M · 1024 px · batch 2
30 epochs on synthetic data
Independent real-world evaluation



business cards · keys · other · paper/notebooks · pens · rulers · scissors · tapes · USB sticks

Synthetic validation confirmed a working baseline





0.953	0.930	0.961	0.814
Precision	Recall	mAP@0.5	mAP@0.5–0.95

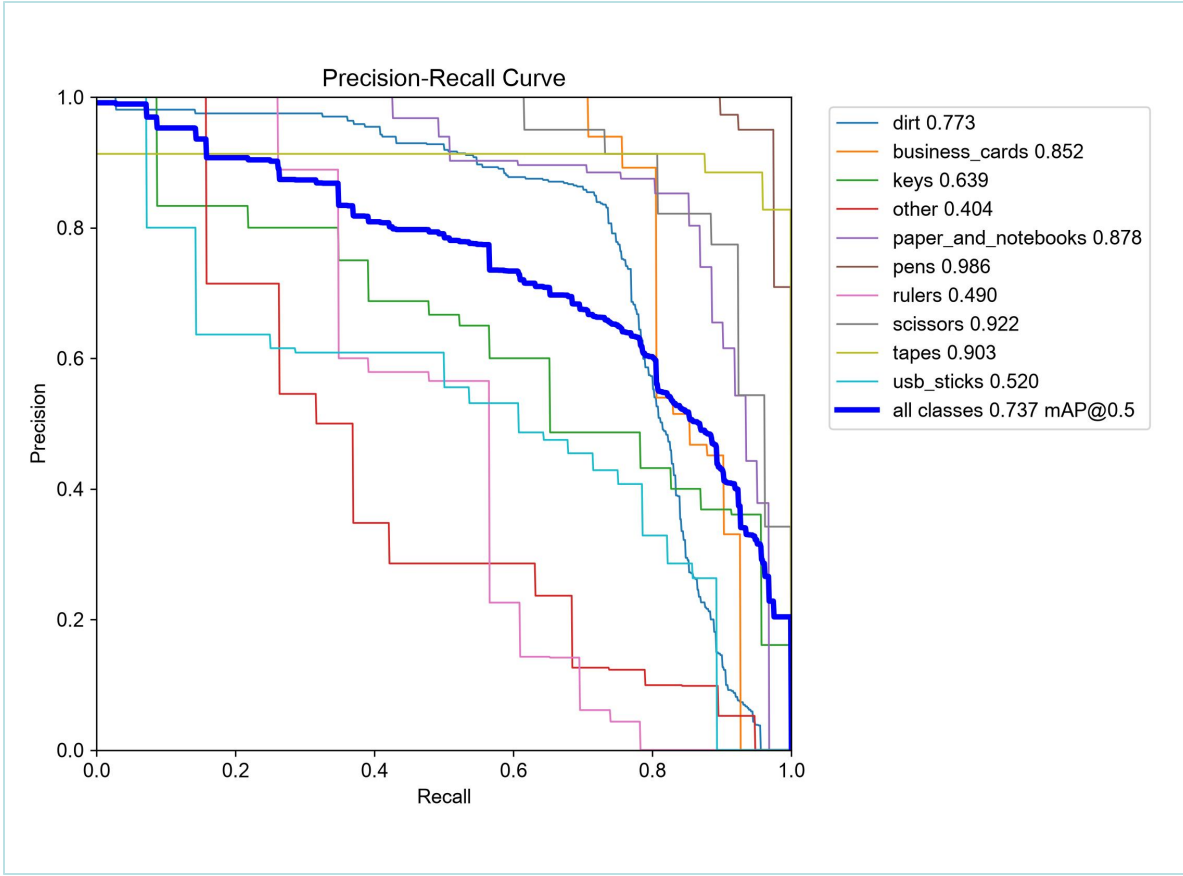
Meaning: the 10-class labels, output head and training pipeline function correctly.

Real images exposed a clear domain gap

	Synthetic validation	Independent real test
Precision	0.953	0.665
Recall	0.930	0.776
mAP@0.5	0.961	0.737
mAP@0.5–0.95	0.814	0.554

mAP@0.5 dropped by 0.224

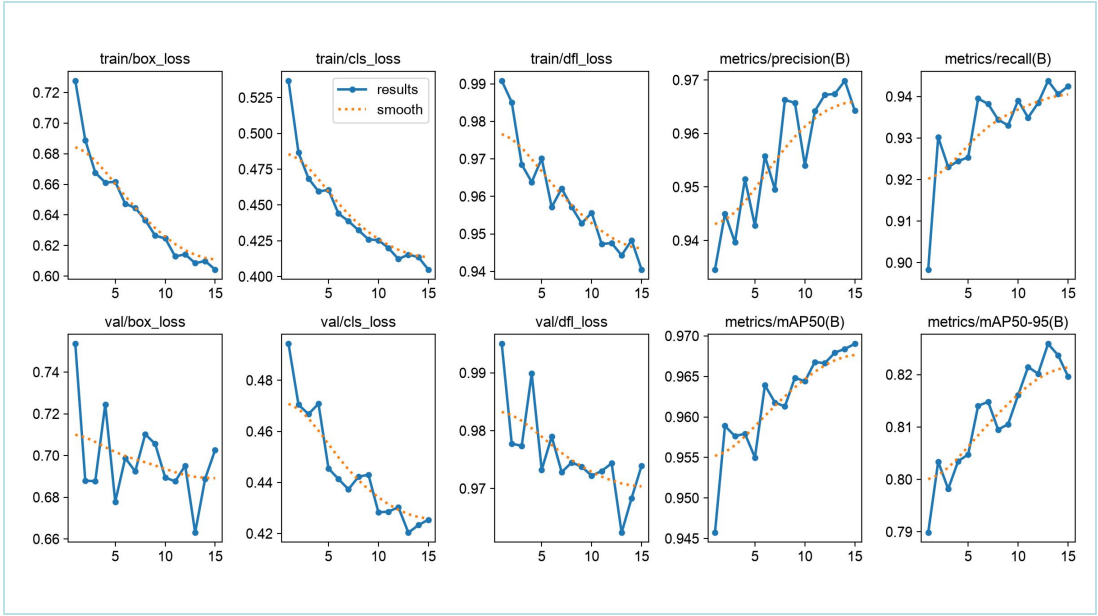
Strong: pens, scissors, paper/notebooks
Weak: other, rulers, USB sticks
Reflections, shadows and floor texture increase false positives



The gap affects both classification and box localization.

Low-rate fine-tuning did not improve real transfer

Modified parameter	Why it was changed
Baseline best.pt	Refine an already converged solution
AdamW	Adaptive steps + decoupled weight decay
$lr_0 = 1 \times 10^{-4}$	Smaller updates preserve learned features
Cosine decay → 1×10^{-5}	Smooth convergence near the optimum
Mosaic disabled	Stabilize layout and localization
15 epochs · patience 5	Bounded refinement; stop early



Synthetic mAP@0.5

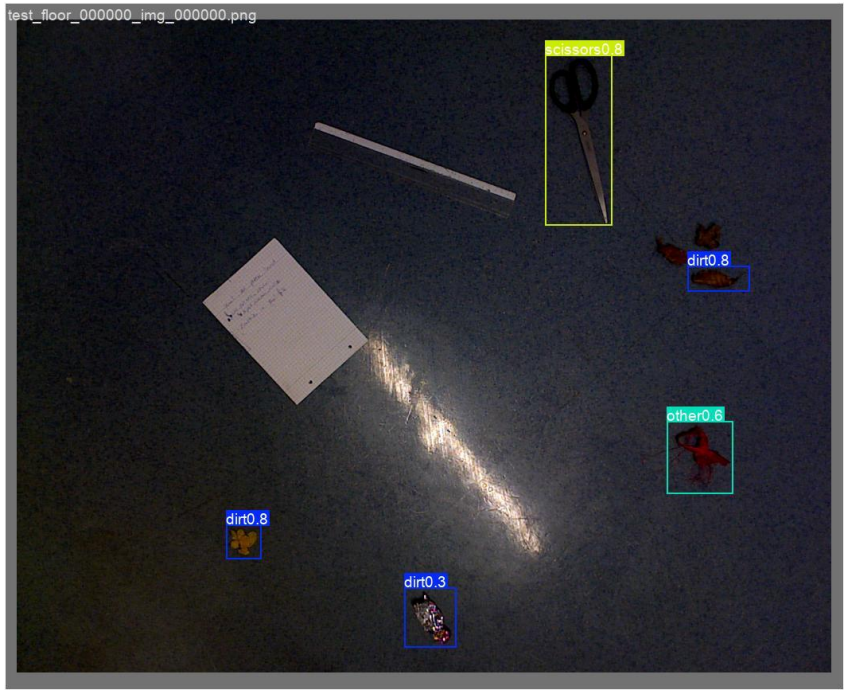
0.961 → 0.968

Real mAP@0.5

0.737 → 0.726

Further optimization fitted the synthetic domain more closely, but did not close the real-world gap.

Further gains require real-world data



Correct detections coexist with missed objects and reflective-floor ambiguity.

Critical discussion & lessons learned

High synthetic scores do not guarantee real-world performance.
More epochs and lower learning rates cannot solve domain mismatch.
Class balance, label definitions and data distribution dominate the remaining error.
The broad "other" class introduces substantial ambiguity.

Outlook and future work

Collect separate real training and validation data.
Add weak classes and difficult negative samples.
Model reflections, shadows, scratches and low light.
Redefine "other" and tune confidence for robot priorities.

A functional multiclass baseline is established; real-domain adaptation is the next step.